

FİZ1951-MUHENDİSLER İÇİN YARIİLETKEN FİZİĞİ ARA-2 FORMÜL KAĞIDI

$$\epsilon_0 = 8.85 \times 10^{-12} \frac{F}{m}$$

$$\mu_0 = 4\pi \times 10^{-7} T \cdot A / m = 1.257 \times 10^{-6} H / m$$

$$q = e = 1.6 \times 10^{-19} C$$

$$h = 6.63 \times 10^{-34} J \cdot s \quad m_0 = 9.11 \times 10^{-31} kg$$

$$k_B = 1.38 \times 10^{-23} J / K \quad 1eV = 1.6 \times 10^{-19} J$$

$$c = 3 \times 10^8 \frac{m}{s}$$

$$\sigma = \frac{ne^2\tau}{m_e^*} \quad \vartheta_t = \sqrt{\frac{3k_B T}{m^*}}$$

$$\frac{1}{\mu_n^T} = \frac{1}{\mu_n^L} + \frac{1}{\mu_n^I}$$

$$\frac{qpV_H}{B_z} = \frac{I_x}{t}$$

$$q(\vec{\vartheta} \times \vec{B}) = q\vec{E}$$

$$qv_s B = qE_{Hall}$$

$$H = \frac{NI}{l}$$

$$M = \chi_M H$$

$$B = \mu_0 H + \mu_0 M$$

$$H = \frac{B}{\mu}$$

$$\mu = \mu_0 \mu_r = \mu_0 (1 + \chi_M)$$

$$V_{Hall} = R_{Hall} \frac{IB}{t}, \quad R_{Hall} = \frac{1}{nq}$$

$$\sigma_p = pq\mu_p \quad \sigma_n = nq\mu_n$$

$$E_{Hall} = v_s B \quad v_s = \frac{I}{nqA}, \quad A = t \cdot W$$

$$R = \frac{(n_r - 1)^2 + k^2}{(n_r + 1)^2 + k^2}$$

$$n_r = \frac{c}{\vartheta}$$

$$\alpha(h\nu) = A(h\nu - E_g)^\gamma$$

$$E = \frac{hc}{\lambda}$$

$$I(d) = I_0 \exp(-\alpha d)$$

$$T = (1 - R^2) \exp(-\alpha d)$$

$$E = \frac{1.24 eV \cdot \mu m}{\lambda(\mu m)}$$

$$E = \frac{1240 eV \cdot nm}{\lambda(nm)}$$

$$\vec{x} = \vec{v}_s \Delta t$$

$$I = \frac{\Delta Q}{\Delta t}$$

$$I = nqA\vartheta_s$$

$$J = \frac{I}{A}$$

$$\vec{J} = nq\vartheta_s$$

$$V = IR$$

$$R = \rho \frac{l}{A}$$

$$\vec{v}_s = \mu \vec{E} = \frac{q\tau}{m} \vec{E}$$

$$\vec{J} = nq\mu \vec{E}$$

$$\sigma = nq\mu = ne\mu$$

$$q = e$$

$$\mu_n = \frac{e\tau_n}{m_n^*}$$

$$\mu_p = \frac{e\tau_p}{m_p^*}$$

$$\vec{J} = \sigma \vec{E} \quad \vec{E} = \frac{V}{l}$$

$$\rho = \frac{1}{ne\mu}$$

$$l = u_{ort} \tau$$

$$u_{th} = l / \tau$$

$$\sigma = 1/\rho \rightarrow S = 1/R$$

$$\vec{F}_e = q\vec{E} = m_0 \vec{a}$$

$$\vec{F}_e = q\vec{E} + q\vec{E}_{Kr} = m^* \vec{a}$$

n^e
↓
Kinn
migi

$n_r - i(k)$ (circled) *konkretne
väärtused*

150k \$ all

↓
Gülar
Köster